

Local Network Discovery & Connectivity Test

Objective

The purpose of this study is to analyze the local network configuration of a computer and verify internet connectivity using basic network diagnostic tools.

Tools Used

Commands used in this study:

- `ipconfig /all`
- `ping`

Step 1 – Local Network Information

The command was executed: `ipconfig /all`

Findings

The computer's network information was examined:

- **IPv4 Address:** 192.168.x.x
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.x.1
- **DNS Servers:** 46.x.x.x, 178.x.x.x
- **DHCP Enabled:** Yes
- **Physical Address (MAC):** XX-XX-XX-XX

* Sensitive network information has been anonymized for security purposes.

Result:

It has been observed that the DNS servers use the ISP's public DNS addresses instead of the local network address (192.168.x.x). This indicates that the router assigns the ISP's public DNS servers to clients via DHCP. The router acts as a gateway between the local network and the ISP infrastructure.

Step 2 – Connectivity Test (IP Address)

The command was executed: `ping 8.8.8.8`

Output

Reply from 8.8.8.8: bytes=32 time=27ms TTL=116

Reply from 8.8.8.8: bytes=32 time=25ms TTL=116

Reply from 8.8.8.8: bytes=32 time=20ms TTL=116

Reply from 8.8.8.8: bytes=32 time=19ms TTL=116

Ping statistics:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Result

ICMP packets were successfully delivered with 0% packet loss.

The average latency was approximately 23 ms, indicating a healthy and stable internet connection.

The TTL value of 116 suggests that packets reached the destination through multiple network hops over the internet.

This test verifies Layer 3 (Network Layer) connectivity to the public internet.

Step 3 – Connectivity Test (Domain Name)

The command was executed: ping google.com

Output

Pinging google.com [192.178.x.x] with 32 bytes of data:

Reply from 192.178.x.x: bytes=32 time=22ms TTL=116

Reply from 192.178.x.x: bytes=32 time=23ms TTL=116

Reply from 192.178.x.x: bytes=32 time=26ms TTL=116

Reply from 192.178.x.x: bytes=32 time=20ms TTL=116

Ping statistics:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Result

This test confirms that DNS resolution is functioning correctly.

The domain name was successfully translated into an IP address, and the target server responded without packet loss.

This verifies connectivity at both the network and DNS levels.

This test validates application-level connectivity using a domain name instead of a direct IP address.

Key Concepts Learned

During this exercise, the following concepts were reinforced:

- Private IP address
- The concept of a default gateway
- The DNS resolution process
- ICMP and the use of ping
- Difference between IP connectivity and DNS resolution

Conclusion

The results confirm that the local network configuration, DNS resolution, and external connectivity are functioning correctly. Basic network diagnostic tools were successfully used to validate end-to-end connectivity.

This study demonstrates the practical use of fundamental network troubleshooting tools used by NOC and Network Engineers.